

DV2627 V24 LP3 Advanced Machine Learning

Blekinge Institute of Technology Project Report

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I. INTRODUCTION

AI-written reviews and comments represent a novel manifestation of synthetic text generated by algorithms mimicking human language patterns. Regrettably, individuals exploit this technology to fabricate deceptive reviews and comments, undermining the integrity of online platforms and consumer trust. Such misuse perpetuates misinformation, distorts public perception, and compromises decision-making processes [9]. Consequently, the proliferation of AI-generated content erodes the credibility of online interactions and imposes profound ramifications on individuals' lives, shaping their opinions, purchasing behaviours, and even societal perceptions. As these AI-generated texts proliferate, robust detection mechanisms become imperative to safeguard the authenticity and trustworthiness of digital discourse.

The recent cyber complaint filed by the movie "Gunturu Kaaram" against "Book My Show" unveils the susceptibility of online platforms to manipulation. Over 70,000 fabricated votes and reviews, primarily with ratings of 0 or 1, skewed the film's ratings artificially. The Film Chamber's subsequent investigation underscores the pressing need for enhanced safeguards [14]. Similarly, nearly half of Amazon's best-selling item reviews lack reliability, with up to 90% unreliability in top-selling categories like clothing, shoes, and jewelry [18].

Addressing the issue of AI-generated reviews and comments is paramount for preserving trust in online platforms, ensuring integrity, and combating misinformation. Detecting and mitigating such content fosters fair competition and cultivates a transparent digital environment where authenticity prevails. The challenge has spurred advancements in detection technologies, propelling science and industry forward. Increased awareness of digital manipulation stimulates ethical discourse on AI usage, driving transparency and responsible practices. This synergy between industry leaders and ethical considerations fuels innovation, shaping a more conscientious digital landscape [11].

In the current research stage, advancements in detection technologies, such as statistical analysis, machine learning, and multimodal approaches, have shown promising results, with accuracy rates exceeding 90% in some cases. However, challenges persist, including keeping pace with evolving AI

capabilities, addressing biases in training data, and navigating ethical concerns. The future of combating AI-crafted content lies in collaboration among researchers, platforms, and policymakers to refine detection methods, tackle ethical challenges, and ensure a genuine online space where AI empowers real voices.

The task involves identifying and mitigating the proliferation of AI-generated reviews and comments, which pose significant challenges to the integrity of online platforms and consumer trust. This problem falls within data mining, where the objective is to detect and combat the fabrication of deceptive content generated by AI mimicking human language patterns. Leveraging exploratory data analysis and machine learning techniques within the CRISP-DM process model [21], the proposed solution entails developing robust detection mechanisms to safeguard the authenticity of digital discourse. However, challenges persist, including keeping pace with evolving AI capabilities, addressing biases in training data, and navigating ethical concerns. The proposed solution emphasises collaboration among researchers, platforms, and policymakers to refine detection methods, tackle ethical challenges, and ensure a genuine online space where AI empowers real voices. Through this approach, the project endeavours to promote integrity and accountability in digital discourse, contributing to a more conscientious and trustworthy online landscape.

AI text detection, which discerns between human and AI-generated content, is pertinent to science by advancing the understanding of AI's influence on digital discourse. This research fosters interdisciplinary collaboration, utilizing exploratory data analysis, machine learning, and ethical considerations to uphold integrity in online communication.

In the broader societal context, AI text detection plays a crucial role in preserving online integrity and combating misinformation. By distinguishing between human and AI-generated content, this technology promotes transparent communication and empowers individuals to make informed decisions, contributing to a more responsible and trustworthy online society.

From an industrial standpoint, AI text detection is instrumental in addressing the proliferation of AI-generated content across digital platforms. By accurately identifying the origin of text, this technology helps businesses maintain the authenticity

and credibility of their online interactions. It aids in combating misinformation, safeguarding consumer trust, and enhancing the competitiveness and reputation of companies in the marketplace.

Ethical considerations are paramount in AI text detection to ensure the responsible development and deployment of AI technologies. By addressing concerns related to digital manipulation, misinformation, and bias, this research fosters ethical discourse and transparency in AI usage. It promotes accountability among stakeholders and advocates for fair and transparent practices, ultimately shaping a more conscientious digital landscape.

The proliferation of AI-generated reviews and comments poses a significant threat to the integrity of online platforms and consumer trust. Consumers rely heavily on online reviews and comments to make informed decisions about products and services. However, the prevalence of deceptive AI-generated content has led to instances where individuals have made purchasing decisions based on false or misleading information, resulting in financial loss and dissatisfaction. This erosion of trust in online platforms and the potential harm caused to consumers motivated me to undertake this study. By investigating the problem of AI-generated comments and reviews, the aim was to contribute to the development of robust detection mechanisms that could safeguard the authenticity of digital discourse and empower consumers to make more informed decisions in the online marketplace.

The project aimed to contribute to the advancement of detection technologies and ethical considerations in AI development and deployment. By addressing the challenges posed by AI-written reviews and comments, the team endeavored to promote integrity and accountability in digital discourse, shaping a more conscientious and trustworthy online landscape.

II. RELATED WORK

After reviewing these papers [12], [13], [15], [16], [20] .Hu, Chen, and Tsung-Yi introduced RADAR, a novel framework for training robust AI-text detectors, outperforming existing methods in detecting AI-generated text. Altunbey Ozbay and Alatas proposed a two-step method for fake news detection on online social media, employing supervised learning algorithms. Smitha and Bharath compared machine learning classifiers for fake news detection, identifying SVM Linear with TF-IDF as the most suitable. Wei et al. developed a robust text detector using contrastive learning, achieving notable results on challenging datasets. Khanam, Alwasel, Sirafi, and Rashid explored various ML approaches, highlighting the effectiveness of the random forest classifier in distinguishing fake news.

While several studies have focused on detecting fake news in articles and social media posts, there appears to be a notable research gap in the detection of AI-generated text in comments and reviews, particularly in the context of movie and product reviews. The existing literature reviewed includes research by Xiaomeng Hu et al., Feyza Altunbey

Ozbay and Bilal Alatas, Smitha and Bharath, and Ran Wei et al., all of which primarily address the detection of fake news in articles and social media content. While these studies contribute valuable insights into the use of machine learning algorithms and text feature extraction techniques for identifying fabricated content, none specifically delve into the detection of AI-generated text in comments and reviews. This highlights an area of research that warrants further exploration, particularly given the growing prevalence of AI-generated content and its potential impact on online discourse, consumer trust and advancements in AI text detection and classification.

III. METHODOLOGY

The methodology employed in this project follows the CRISP-DM model, renowned for its cost-efficiency, reliability, reproducibility, manageability, and speed in data mining projects [21]. This model comprises six phases:

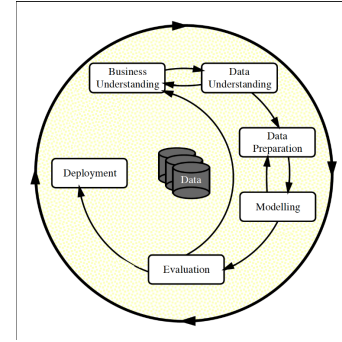


Fig. 1. CRISP-DM model [21]

A. Business Understanding:

In the Business Understanding phase, the project was prioritized for a comprehensive comprehension of the ramifications of AI-generated reviews and comments on online platforms, as outlined in the introduction. The project delved into understanding the imperative to preserve trust in digital discourse, combat misinformation, and ensure integrity. By recognizing the detrimental impact of deceptive AI-generated content on consumer trust and decision-making processes, the project aimed to develop robust AI text detection mechanisms and an advanced machine learning system. This phase laid the groundwork for aligning project goals with the overarching objective of promoting integrity and accountability in digital communication.

B. Data Understanding:

In the Data Understanding phase, the project was conducted a thorough exploration of datasets encompassing AI-generated reviews and comments. This phase involved scrutinizing multiple datasets to pinpoint four that were deemed most relevant and suitable for the project's objectives:

- 1) "Detect- AI Generated VS Student Generated Text":
This dataset comprising 1103 records and two columns:

"Text" and "Label." The "Text" column contains text authored by either AI or human (student), while the "Label" column indicates the origin of the text as either AI-generated or human-generated. This dataset is formatted as a CSV file and contains no missing values or duplicates. Each text entry ranges from 22 to 100 characters in length, with an average length of 56 characters. [3].

- 2) "AI Vs Human Text": This dataset comprises approximately 500,000 essays, sourced from multiple channels and meticulously curated to eliminate duplicates. This dataset is in CSV format, containing 487,235 records and two columns: 'text' and 'Generated by AI'. The 'text' column consists of unique textual data, while the 'Generated by AI' column categorizes each entry as either AI-generated (1) or human-written (0). There are no missing values or duplicates in the dataset. The length of the text varies from 1 to 18,322 characters, with the majority of entries having a length of 1,974 characters, serving as a comprehensive foundation for further analysis and model development in alignment with the project's objectives. [1].
- 3) "Amazon Product Reviews": This dataset comprises over 568,000 consumer reviews from various Amazon products. The dataset, available in CSV format, contains 10 attributes including Id, ProductId, UserId, Profile-Name, HelpfulnessNumerator, HelpfulnessDenominator, Score, Time, Summary, and Text. For this project, the focus was primarily on the "Text" attribute, which represents the actual review content. This attribute alone contains 393,579 unique values, with no missing values but has some duplicates. The review texts ranged from a minimum length of 12 to a maximum length of 21,409 characters, with an average length of 126 characters. Understanding these surface properties of the data laid a solid foundation for identifying, collecting, and analyzing datasets essential for achieving the project's objectives. [2]. This was an unlabeled dataset.
- 4) "IMDB Dataset of 50K Movie Reviews": This dataset comprises around 50,000 movie reviews specifically designed for binary sentiment classification. The dataset contains two columns: "review" and "sentiment," with 49,582 unique values in the "review" column and two unique values ("positive" and "negative") in the "sentiment" column. Only the "review" column was utilized for this project, as it contains textual reviews of movies. The dataset is stored in CSV format and consists of 49,582 records. While there were no missing values or duplicates. Additionally, the team noted that the length of the text ranged from a minimum of 32 to a maximum of 13,704 characters, with a majority of the strings having a length of 658 characters. [4]. This was an unlabeled dataset.

The motivation behind utilizing these datasets for this project was encompassing AI-generated reviews, student es-

says, Amazon product reviews, and movie critiques, offering a rich landscape for comprehensive analysis. By leveraging these datasets, the project aims to develop robust detection mechanisms to combat the proliferation of AI-generated content.

C. Data Preparation:

In the Data Preparation phase, refine the datasets to ensure their readiness for modeling purposes. This phase was divided into two subphases: data preprocessing [7] and exploratory data analysis (EDA) [19]. Data preprocessing techniques were employed to enhance the quality of the data with 11 different techniques. Additionally, exploratory data analysis (EDA) was conducted to extract important insights, identify patterns, and discern relationships within the datasets, laying a solid foundation for subsequent modeling efforts.

1) *Data Preprocessing*: Data preprocessing involves a set of techniques used to clean, transform, and enhance raw data to ensure its quality and suitability for analysis and modeling tasks. In this project, 11 data preprocessing techniques was employed [7]. They are:

- 1) *Data Cleaning*: Data cleaning involves removing HTML tags and special characters from the dataset. The motivation for using data cleaning was to enhance data quality and ensure compatibility with analysis algorithms. This step was justified as it improved the accuracy and reliability of subsequent analyses.
- 2) *Data Integration*: Data Integration merged selected datasets into labeled and unlabeled categories, for optimizing analysis. By combining the selected datasets, the project maximized data utilization, enhancing insights from diverse sources. The motivation behind this approach as it ensured a comprehensive understanding of the data landscape, improving the project's effectiveness in achieving its objectives.
- 3) *Missing Values Handling*: In this step, all the records with missing values are removed. The motivation for using Missing Values Handling is to ensure the integrity and accuracy of the dataset. Justification for using this technique lies in the need to eliminate incomplete or unreliable data, which could otherwise compromise analysis results.
- 4) *Handling Duplicates*: In this step, all records with duplicates are removed, enhancing data integrity. This ensures accuracy in analysis and modeling, motivating the need for clean data. Justification for using this approach lies in improving model performance by eliminating redundancy and potential biases caused by duplicate entries.
- 5) *Text Normalization*: Text Normalization involves converting all records into lowercase. This step was implemented to ensure consistency and simplify subsequent analyses. The motivation for using Text Normalization was to standardize the text data, facilitating easier comparison and reducing the complexity of text processing

tasks. Justification for its use lies in its effectiveness in enhancing data consistency and interpretability.

- 6) **Stopwords Removal:** In this step, all stopwords were eliminated from the text using the NLTK library. This process was motivated by the desire to enhance the quality of textual data by eliminating common words that do not contribute to the analysis. Justification for employing this technique lies in its effectiveness in improving the accuracy of natural language processing tasks.
- 7) **Lemmatization:** It is a text preprocessing technique, was applied to convert words into their base forms using the WordNet corpus. The motivation for using Lemmatization stemmed from its ability to standardize text data, while its justification lay in enhancing the accuracy of subsequent modeling tasks.
- 8) **Handling Emoji and Emoticons:** This step was involved to remove all emojis and emoticons from the text data since ML algorithms and analysis cannot handle them. The motivation for using this step was to ensure that the data remains compatible with the subsequent analytical processes. Justification for this step lies in enhancing the data quality for accurate modeling.
- 9) **Data Encoding:** Data Encoding refers to the process of converting categorical label data into standard binary format, assigning 0 for human-written and 1 for AI-generated text. The motivation for using Data Encoding was to standardize the label data for consistency in model training. Justification for its use lies in ensuring uniform representation for effective model learning.
- 10) **Tokenization:** Tokenization involves splitting text data into individual tokens. In this step, all the text data was tokenized into tokens. The motivation for using tokenization lies in its ability to break down text into smaller units for analysis. Justification for using tokenization stems from its essential role in preprocessing textual data for further analysis.
- 11) **Handling Imbalanced Data:** In this step, all the imbalanced data was addressed. As a result, the data was found to be balanced, eliminating the need for further action. The motivation for using Handling Imbalanced Data was to ensure fair representation of classes. Justification for its use lay in improving model performance and accuracy.

2) *Exploratory Data Analysis (EDA):*

- 1) **Data Information:** The labelled dataset comprised 487,898 entries with two columns: "Text" and "Label," denoting the text content and corresponding labels, respectively. The unlabelled dataset consisted of 443,079 entries, with a single column containing textual data. Motivated by the need to analyze both labelled and unlabelled data, this approach justified comprehensive exploration and model development for robust classification.
- 2) **Text Length Analysis:** Text Length Analysis involves

examining the length of textual data within a dataset. The motivation for using Text Length Analysis lies in understanding the distribution of text lengths, which can reveal insights into the complexity and depth of language usage. Justification for using Text Length Analysis rests on its ability to inform feature engineering and modeling decisions based on text length characteristics.

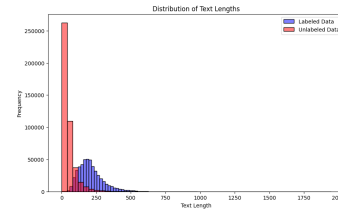


Fig. 2. Word Frequency Analysis:

- 3) **Average Word Length Analysis:** The "Average Word Length" is defined as the mean number of characters per word in a given dataset. In this project, the average word length was found to be 5.86 for labeled data, 5.46 for unlabeled data, and 5.79 for combined data. Motivation for using "Average Word Length" lies in its ability to provide insights into the linguistic complexity of textual data. Justification for its use stems from its relevance in understanding language patterns and aiding in feature engineering for text-based machine learning models.
- 4) **Word Frequency Analysis:** Word Frequency Analysis involves counting the occurrences of words within a given text to identify the most common terms. The motivation for using Word Frequency Analysis lies in its ability to unveil prominent themes and patterns within textual data. Its justification stems from its effectiveness in extracting key insights and understanding the underlying context of the text.
 - Most Common Words - Labeled Data at Fig.8
 - Most Common Words - Unlabeled Data at Fig.9
 - Most Common Words - Combined Data at Fig.10
 - Most Common Words Generated By AI Bots at Fig.11
 - Most Common Words Generated By Humans at Fig.12
- 5) **Word Cloud:** The word cloud, a visual representation of text data, provides a concise summary of frequently occurring words. Its application served as a motivational tool, offering a visually compelling insight into key themes within textual data. Justification for its use stemmed from its ability to highlight prominent keywords, facilitating data interpretation and analysis.
 - Word Cloud for Labelled data at Fig.13
 - Word Cloud for Unlabelled data at Fig.14

D. *Modeling:*

- 1) **Text Extraction:** The Text Extraction phase involved extracting texts and labels from the dataset, employing

the eval function to convert string representations of lists to actual lists, then extracts texts and labels into separate lists. This approach was motivated by the need to efficiently process and organize textual data for subsequent analysis. Justification for employing Text Extraction lies in its effectiveness in preparing the dataset for modeling tasks.

- 2) Text Data To Sequences: This process was involved in tokenizing the combined text data using Tensorflow Tokenizer and converting the tokenized text data to sequences of integers. The motivation for employing this approach was to efficiently preprocess text data for machine learning tasks. Justification for its use stemmed from its effectiveness in numerical representation of textual information.
- 3) Padding: Padding is the process of adding dummy values to sequences to ensure uniform length, resulting in sequences of equal length. This ensures compatibility for modeling algorithms. The motivation for using padding is to handle variable-length input data. Justification lies in its ability to facilitate batch processing and enhance computational efficiency.
- 4) Data Splitting: This step was involved in partitioning the dataset into training, testing, and validation sets with a ratio of 60:20:20, respectively. This was motivated by the need to assess model performance accurately while ensuring generalizability. Justification for this approach lies in its ability to provide robust evaluation metrics and prevent overfitting.
- 5) Model selection: The process of choosing the most suitable algorithm or method for a given task, was pivotal in this project. After careful consideration and evaluation, the selected model to perform was random forest.

The motivation for using random forest stemmed from its versatility, robustness, and proven effectiveness in handling both classification and regression tasks. Its ability to mitigate overfitting, handle large datasets, and provide valuable insights into feature importance made it an ideal choice for detecting AI-generated reviews and comments [6].

Justification for using random forest was further supported by its ability to handle high-dimensional data, deal with imbalanced datasets, and deliver reliable performance across various domains. Additionally, its ease of implementation and interpretability aligned well with the project's objectives, facilitating the development of a robust detection mechanism to safeguard the authenticity of digital discourse [6].

Overall, the decision to employ random forest for model selection was driven by its strong track record, versatility, and suitability for addressing the challenges posed by AI-generated content in online platforms.

- 6) ML Modeling: In this ml modeling step, the project loaded the random forest model, named Baseline Model, and trained it using the available training data. After

training, the model was saved for future use. Subsequently, pseudo-labelling was performed on the unlabelled data using the Baseline Model by making predictions. The pseudo-labelled data was then utilized to retrain the Baseline Model, resulting in the creation of a combined data model named combined-model.

Following model validation using the validation data, both models exhibited an accuracy of 81%. Given their identical performance, a comprehensive evaluation of both models was conducted. Utilizing model evaluation metrics such as confusion matrices, accuracy, precision, recall, F1 Score, and AUC metrics with the test data, the evaluation yielded the following results:

Evaluation Metric	Baseline Model Score	Pseudo Labeled Model Score
Accuracy	0.87	0.87
Precision	0.89	0.86
Recall	0.75	0.77
F1- Score	0.81	0.81
AUC-ROC	0.95	0.94

TABLE I
RESULTS COMPARISON

The comparison of results highlights the superior performance of the baseline model over the combined model in several key aspects. With fewer false positives and higher precision, the baseline model demonstrates better accuracy and classification capability. Despite similar F1 scores, the baseline model's higher AUC underscores its effectiveness in distinguishing between positive and negative instances. Overall, these findings affirm the baseline model's superiority in the classification task.

The other reasons for the baseline model likely outperforms the model retrained with pseudo-labeled data due to several factors such as Pseudo-labeled data introduces noise, risking overfitting, model complexity, unlabeled data quality and domain shifts. Additionally, it may increase model complexity and suffer from lower-quality unlabeled data. These challenges can degrade the retrained model's performance compared to the baseline, which relies solely on clean, labeled data for training, resulting in superior generalization capabilities. So, the Baseline model was selected to use for the further steps.

- 7) Hyperparameter Tuning: Hyperparameter tuning involves optimizing the settings or configurations of a machine learning algorithm to enhance its performance. In this project 7 parameters were selected to perform hyperparameter tuning [17]. They are:

- n_estimators: The parameter "n_estimators" was defined as the number of trees in the random forest classifier. Hyperparameter tuning determined the best value as 300 among 100,200 and 300. Motivation for using "n_estimators" was to optimize model performance. Justification lies in enhancing model robustness [5].

- **max_depth:** The "max_depth" parameter represents the maximum depth of the decision tree in a random forest classifier. Hyperparameter tuning yielded the best value of 20 among None, 10, 20 for "max_depth." The motivation for using "max_depth" is to control the depth of the trees, which can help prevent overfitting. Justification for its use lies in its ability to optimize the model's performance by limiting tree complexity [5].
- **min_samples_split:** The parameter 'min_samples_split' specifies the minimum number of samples required to split an internal node. Hyperparameter tuning determined the best value as 2 among 2, 5, 10. Using it motivates improved model performance and generalization. Justification relies on optimizing decision boundaries [5].
- **min_samples_leaf:** The hyperparameter "min_samples_leaf" was optimized through tuning, yielding the best value of 2 among 1, 2, 4. This choice was motivated by the desire to control overfitting. Justification lies in achieving better generalization [5].
- **Bootstrap:** The bootstrap parameter, which determines whether bootstrap samples are used when building trees in a random forest classifier, was tuned, resulting in the best value being False. This choice was motivated by the desire to optimize model performance while minimizing overfitting. Justification for this decision lies in the potential to improve the generalization ability of the model by reducing variance [5].
- **Class_Weight:** The class_weight hyperparameter that assigns weights to classes within the dataset for the random forest classifier was tuned, with the best value found to be 'balanced' among None, 'balanced', 'balanced_subsample'. This choice was motivated by the need to address class imbalance. Justification for this selection lies in its ability to adjust the class weights to better handle skewed datasets [5].
- **Criterion:** The criterion, a hyperparameter for the random forest classifier, refers to the measure used for splitting nodes. In this study, hyperparameter tuning identified 'entropy' as the best criterion, chosen for its superior performance. The motivation for using entropy lies in its ability to capture more complex decision boundaries, enhancing model robustness and accuracy. Justification for its selection is grounded in empirical evidence from the hyperparameter tuning process, which demonstrated its effectiveness in optimizing model performance [5].

By aggregating the optimal values from each selected parameter, a new random forest model, termed the tuned model, was constructed using combined data and trained accordingly. Subsequently, k-fold cross-validation was

executed with $k = 5$ on validation data, resulting in the tuned model achieving an accuracy of 84%, surpassing the Baseline model and combined model by 3%. Hyperparameter tuning played a pivotal role in this project, facilitating the optimization of the random forest classifier's performance through systematic exploration of diverse parameter configurations. The decision to employ hyperparameter tuning was motivated by the imperative to enhance model performance and generalization across key parameters such as n_estimators, max_depth, and min_samples_split. This approach's justification lies in its efficacy in fine-tuning model parameters to optimize decision boundaries, mitigate overfitting, and enhance overall model generalization. Thus, this systematic approach ensures that the model is finely tailored to the dataset, culminating in superior performance.

E. Evaluation

Model evaluation is the process of assessing the performance and effectiveness of a trained machine learning model using various metrics and techniques [8]. It involves analyzing how well the model performs on unseen or test data compared to its performance on training data. The choice of evaluation metrics for this project was:

- **Accuracy:** The accuracy metric measures the proportion of correctly classified instances by a random forest classifier. Motivation lies in evaluating model performance [8].
- **Precision:** The precision metric measures the ratio of correctly predicted positive observations to the total predicted positives. Motivation for using precision lies in its ability to assess the model's accuracy in identifying positive cases [8].
- **Recall:** Recall, also known as sensitivity, measures the proportion of actual positive cases that were correctly identified. Motivated by the need to accurately capture positive instances [8].
- **F1-score:** The F1-score is a metric that combines precision and recall, providing a balance between them. The motivation for using the F1-score lies in its ability to assess the model's overall performance, accounting for both false positives and false negatives.
- **ROC curve analysis:** ROC curve analysis is a method used to evaluate the performance of a classifier. Its motivation is to assess and visualize classification performance [8].
- **Confusion Matrix Visualization:** Confusion matrix visualization is a graphical representation that illustrates the performance of a random forest classifier. The motivation for using this visualization tool lies in its ability to provide a clear overview of classification performance, aiding in the interpretation of model results [8].

As a part of the evaluation the tuned model was under gose Explainable AI (XAI). Explainable AI (XAI) elucidates how AI models make decisions, enhancing transparency and trust in their outcomes [10]. The choice of explainable AI techniques for this project was:

- **LIME:** LIME (Local Interpretable Model-agnostic Explanations) is a technique used to explain the predictions of machine learning models. The motivation for using LIME lies in its ability to provide interpretable explanations for individual predictions. Justification for using LIME stems from its model-agnostic nature and effectiveness in understanding complex model behavior [10].
- **Feature Importance Analysis:** Feature Importance Analysis refers to the process of determining the most significant features in a dataset. Motivation for using this technique lies in understanding which features contribute most to model performance. Justification rests on improving model interpretability and identifying key predictors [10].
- **Partial Dependence Plots (PDPs):** Partial Dependence Plots visualize the relationship between a feature and the predicted outcome, aiding interpretation. The motivation and justification lie in enhancing model explainability [10].

The Results of the selected evaluation metrics and explainable AI techniques were discussed in the further sections.

F. Deployment

In the Deployment phase, the project's findings and developed models were integrated into practical applications to combat the proliferation of AI-generated content. Leveraging the insights gained from data analysis and model development, the team implemented robust detection mechanisms into online platforms and digital environments. The deployed solutions aimed to uphold the authenticity of digital discourse and empower users to navigate online interactions with confidence.

IV. RESULTS AND ANALYSIS

A. Model Evaluation Metrics

- **Accuracy:** 0.85 indicates model performance which measures how often the tuned model is correct. Higher accuracy indicates higher correct predictions.
- **Precision:** 0.89 which measures the proportion of true positive predictions. Higher precision indicates fewer false positives.
- **Recall:** 0.68 which indicates the proportion of actual positives correctly identified, enhancing model performance.
- **F1 Score:** 0.77 which measures model accuracy considering precision and recall. It balances between precision and recall.
- **AUC-ROC Score:** 0.94 which indicates high model performance in binary classification, measuring the area under the ROC curve.

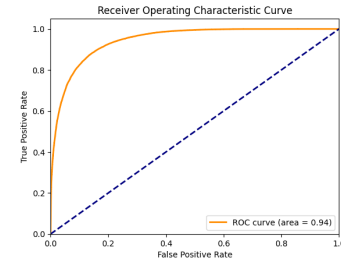


Fig. 3. ROC Curve

- **Confusion Matrix Visualization:** Confusion matrix visualization is a graphical representation that illustrates the performance of a random forest classifier

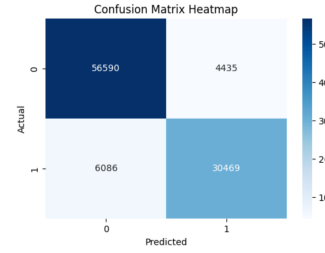


Fig. 4. Confusion Matrix Visualization

B. Explainable AI Techniques

- **LIME:** The below plot shows the LIME results which are used to interpret the predictions of a machine learning model and understand the factors influencing its decisions.

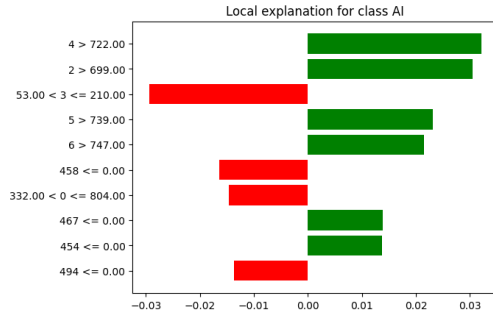


Fig. 5. LIME

- **Feature importance analysis:** The below plot shows the feature importance analysis results which are used to identify the most influential features in a model, aiding in understanding which variables contribute the most to the model's predictions.

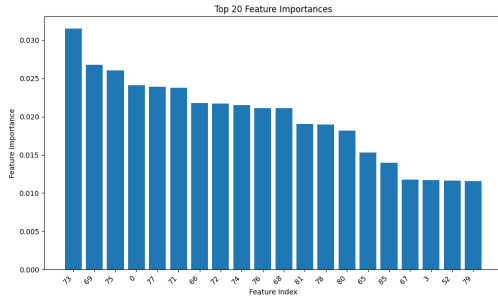


Fig. 6. Feature importances

- **Partial Dependence Plots (PDPs):** The below plot shows the PDPs results which are used to visualize and understand the relationship between a feature and the predicted outcome in machine learning models. They help in interpreting the impact of individual features on the model predictions while holding other features constant.

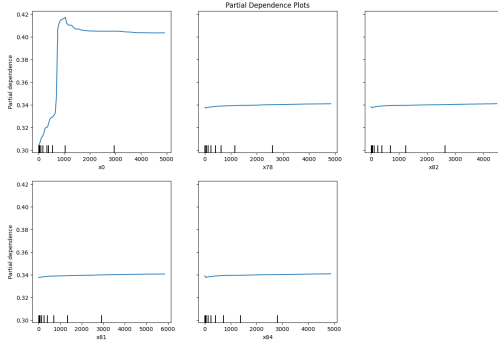


Fig. 7. Partial Dependence Plots

V. DISCUSSION

RQ1: What insights and patterns emerge from the application of Data Mining techniques and Exploratory Data Analysis (EDA) on the collected data regarding AI-generated and human-written reviews and comments?

The application of Data Mining techniques and Exploratory Data Analysis (EDA) on the collected data regarding AI-generated and human-written reviews and comments revealed several insights and patterns. Through thorough exploration, the team analyzed both labeled and unlabeled datasets, totaling 487,898 and 443,079 entries, respectively. Text Length Analysis showcased that most text data fell within the range of 0 to 250 characters, with an average word length of 5.86 for labeled data, 5.46 for unlabeled data, and 5.79 for combined data. Word Frequency Analysis uncovered common terms such as "student" and "like", indicating prevalent themes. Additionally, Word Cloud visualizations provided concise summaries of frequently occurring words, facilitating the interpretation and analysis of key textual themes. These findings contribute to a comprehensive understanding of the dataset, guiding further model development and classification

efforts.

RQ2: To what extent does the developed Advanced Machine Learning system accurately classify reviews and comments as AI-generated or human-written, considering different ML algorithms and techniques?

The developed Advanced Machine Learning system demonstrates commendable performance in accurately classifying reviews and comments as AI-generated or human-written. With an accuracy of 0.85, the model showcases a high level of correctness in its predictions. The precision of 0.89 indicates a strong ability to minimize false positives, while the recall of 0.68 signifies the model's effectiveness in identifying actual positives. Furthermore, the F1 Score of 0.77 reflects a balanced accuracy considering both precision and recall. Notably, the AUC-ROC Score of 0.94 highlights the model's robust performance in binary classification tasks, underscoring its efficacy in distinguishing between AI-generated and human-written content across different ML algorithms and techniques.

VI. LIMITATIONS

- 1) The project utilized data from only four limited datasets.
- 2) This project exclusively employed the Random Forest algorithm, although numerous other machine learning algorithms were accessible.
- 3) This project exclusively utilized machine learning algorithms, although deep learning algorithms such as Artificial Neural Networks (ANN) were also accessible.

VII. CONCLUSIONS

Based on the results obtained from the application of Data Mining techniques and Exploratory Data Analysis (EDA) on the collected data, it can be concluded that the developed Advanced Machine Learning system demonstrates commendable performance in accurately classifying reviews and comments as AI-generated or human-written. The insights and patterns revealed through thorough exploration of the dataset provide a comprehensive understanding of the textual themes and common terms, which can guide further model development and classification efforts. The preprocessing steps, such as text normalization, stopwords removal, and handling imbalanced data, have contributed to improving the accuracy and performance of the model. Additionally, the ethical considerations addressed in the study promote transparency and accountability in AI text detection, ultimately shaping a more conscientious digital landscape.

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APPENDIX

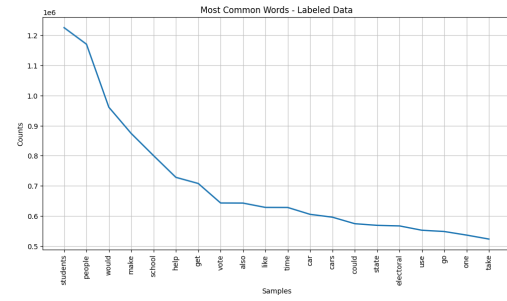


Fig. 8. Most Common Words - Labeled Data

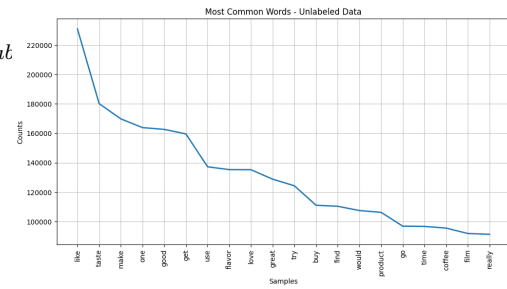


Fig. 9. Most Common Words - Unlabeled Data

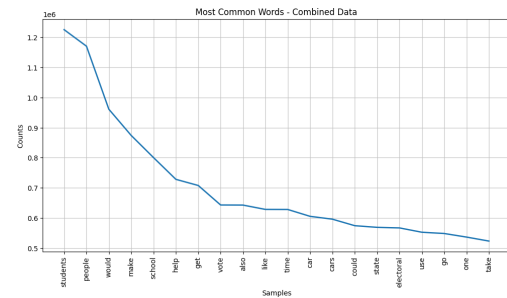


Fig. 10. Most Common Words - Combined Data

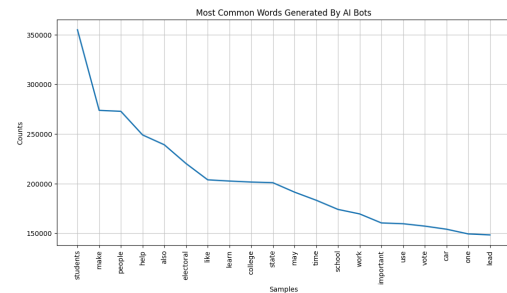


Fig. 11. Most Common Words Generated By AI Bots

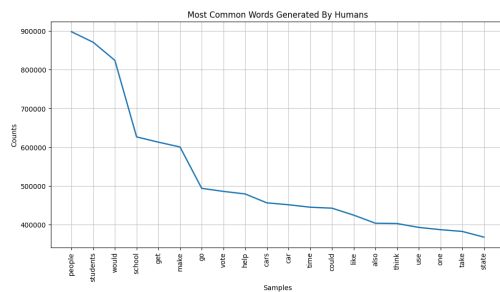


Fig. 12. Most Common Words Generated By Humans

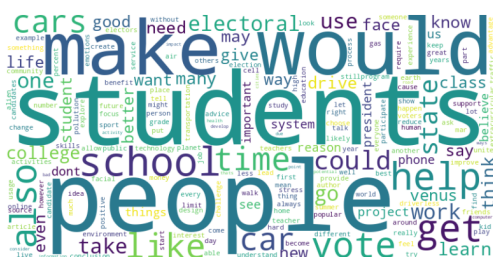


Fig. 13. Word Cloud for labelled



Fig. 14. Word Cloud for Unlabelled